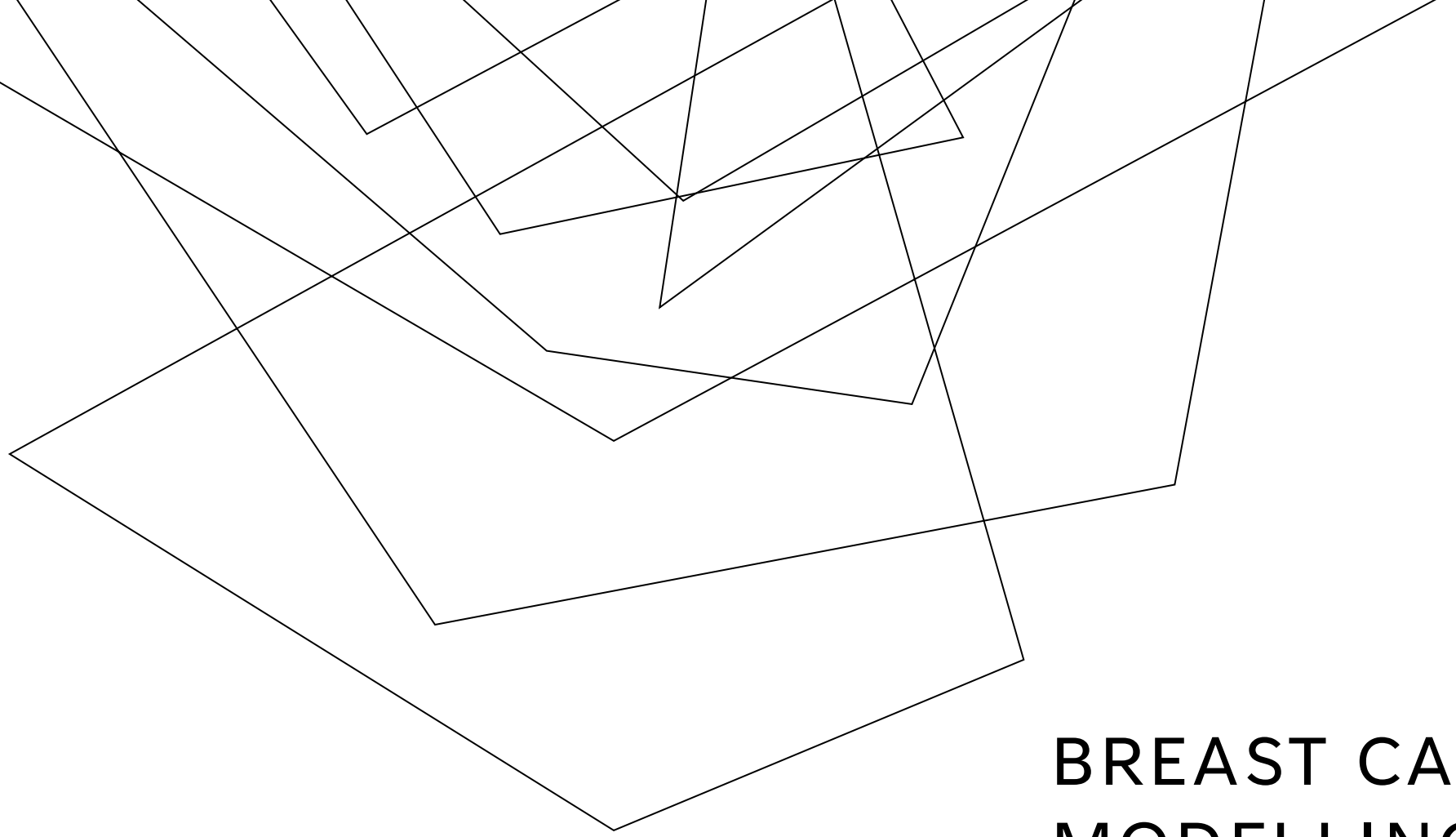


BREAST CANCER: MODELLING AND CLASSIFICATION

Brandon Chung, Jiaxin Zheng,
Andreina Arias, and Stephanie Chiang

DATA 621 – Fall 2025

AGENDA

Abstract

Introduction

Literature Review

Methodology

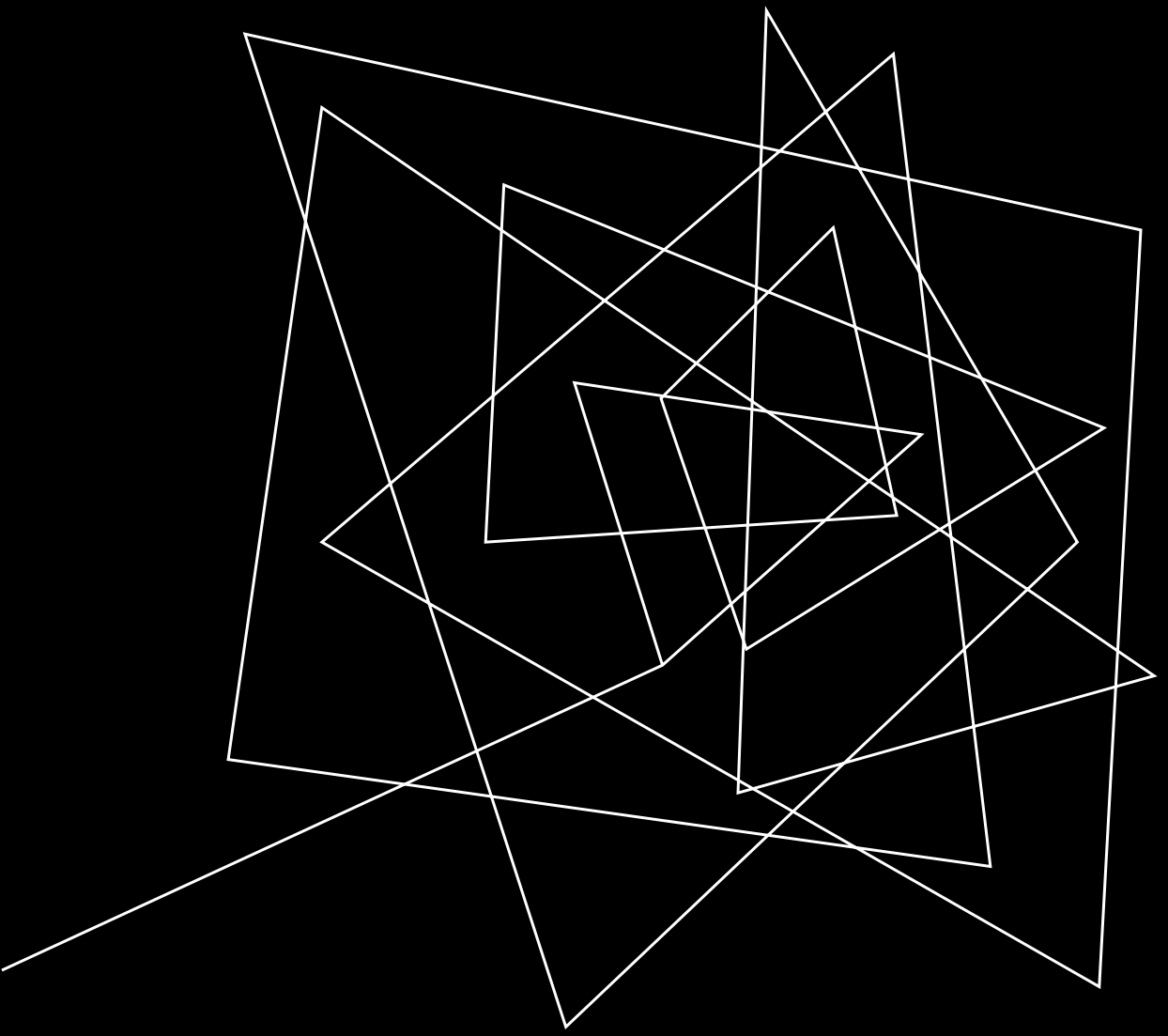
Results

Conclusion

References

ABSTRACT

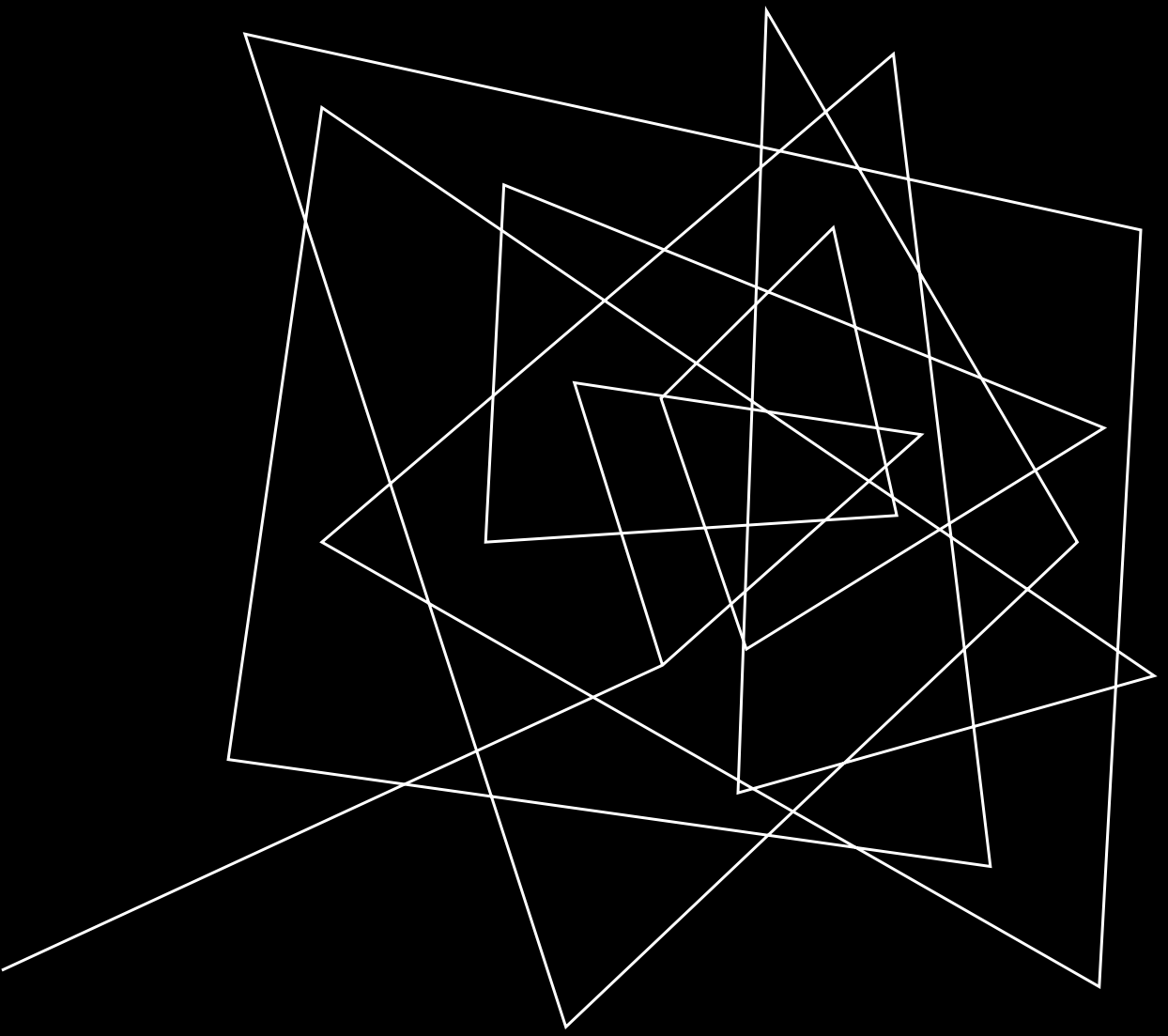
- Investigated whether statistical and machine learning methods can effectively predict whether a breast mass is benign or malignant using radiological measurements; machine learning can potentially reduce unnecessary biopsies if it can make these predictions
- Applied several statistical classification techniques like logistic regression, random tree, and decision tree to model the relationship between radiologic features and diagnostic outcomes.
- Model performance was evaluated using accuracy, ROC curve (AUC), and sensitivity.
- Results demonstrated properly tuned statistical models can achieve high predictive performance, indicating that radiology features contain significant information to distinguish benign and malignant tumors



INTRODUCTION

WHY BREAST CANCER?

- Breast cancer is the most commonly diagnosed cancer among women worldwide (Ferlay et al., 2024)
- Early-stage detection is associated with cure rates as high as 90%, whereas diagnosis in intermediate stages reduces cure rates to 50–80%.
- By examining statistical relationships among tumor features, we can use machine learning modeling methods to provide a transparent and interpretable framework to support clinical decision-making and enhance diagnostic accuracy.



LITERATURE REVIEW

LITERATURE REVIEW

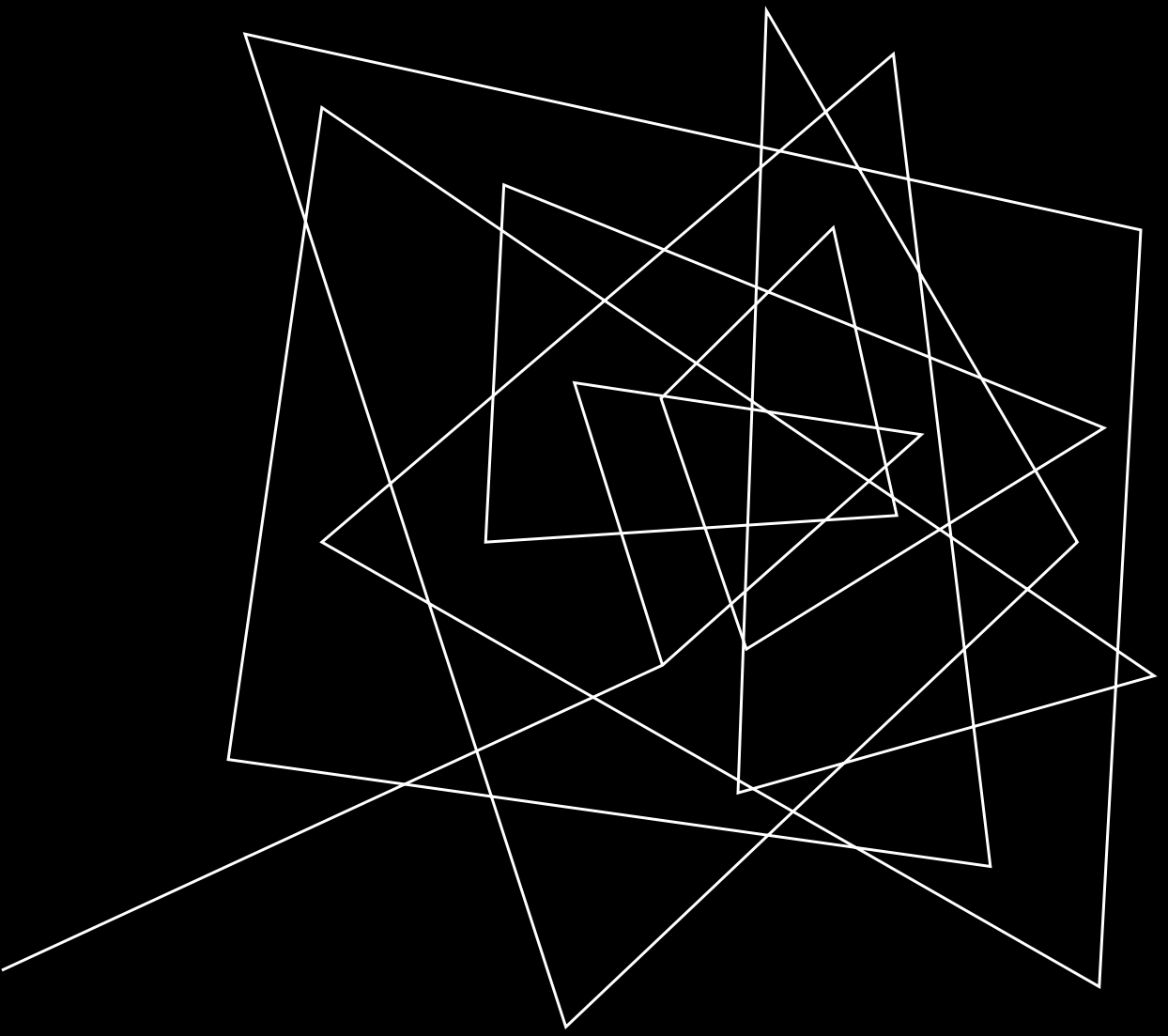
- **Hartmann et al. (2006):** Demonstrated that tumor characteristics (size, histology, hormone-receptor status) strongly influence treatment effectiveness, recurrence, and survival, highlighting the clinical importance of accurate tumor classification.

- **Li et al.:** Showed that breast cancer subtypes have distinct risk-factor profiles, reinforcing that breast cancer is biologically heterogeneous and must be modeled and analyzed at the subtype level.

•**Hassan et al. (2023):** In a comparison between multiple models, Random Forest achieved the highest accuracy (90.86%) when paired with or without LASSO feature selection, outperforming KNN and MLP in breast cancer detection.

•**Duan et al. (2024):** Across multiple diagnostic and metastasis-prediction models, Random Forest again led performance with 93.6% accuracy, an F1-score of 88.9%, and an AUC of 91.3%.

•**Faisal et al. (2018):** Gradient-Boosted Tree was the top performer (90% accuracy) among several ML classifiers for early-stage breast cancer diagnosis



METHODOLOGY

OUR APPROACH

The Data

Consisting of 569 observations with 30 numeric features that describe characteristics of the cell nuclei of breast tumors, such as:

- radius
- texture
- perimeter
- area

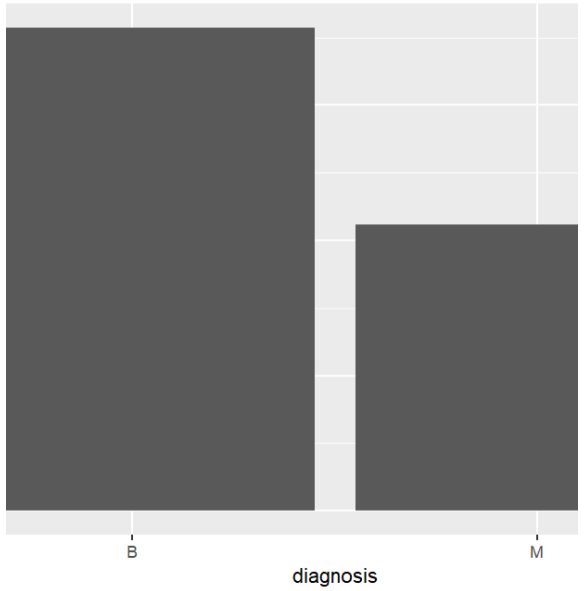
Models

To classify tumors as benign or malignant based on their physical characteristics, we implemented a multi-stage modelling approach including:

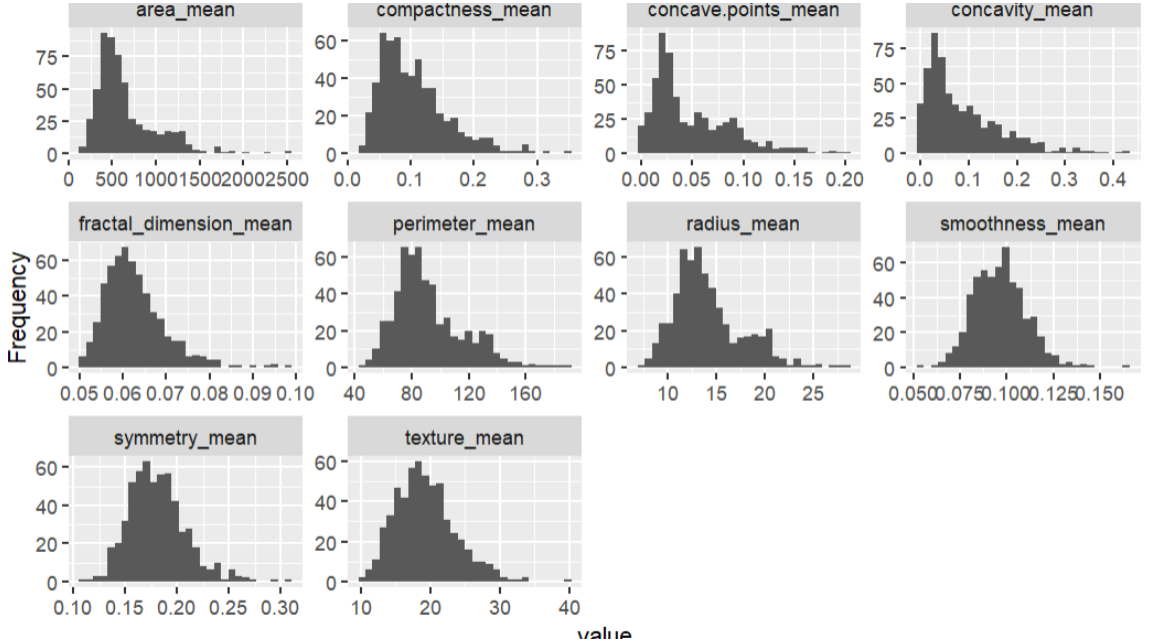
- Logistic Regression
- Decision Tree
- Random Forest

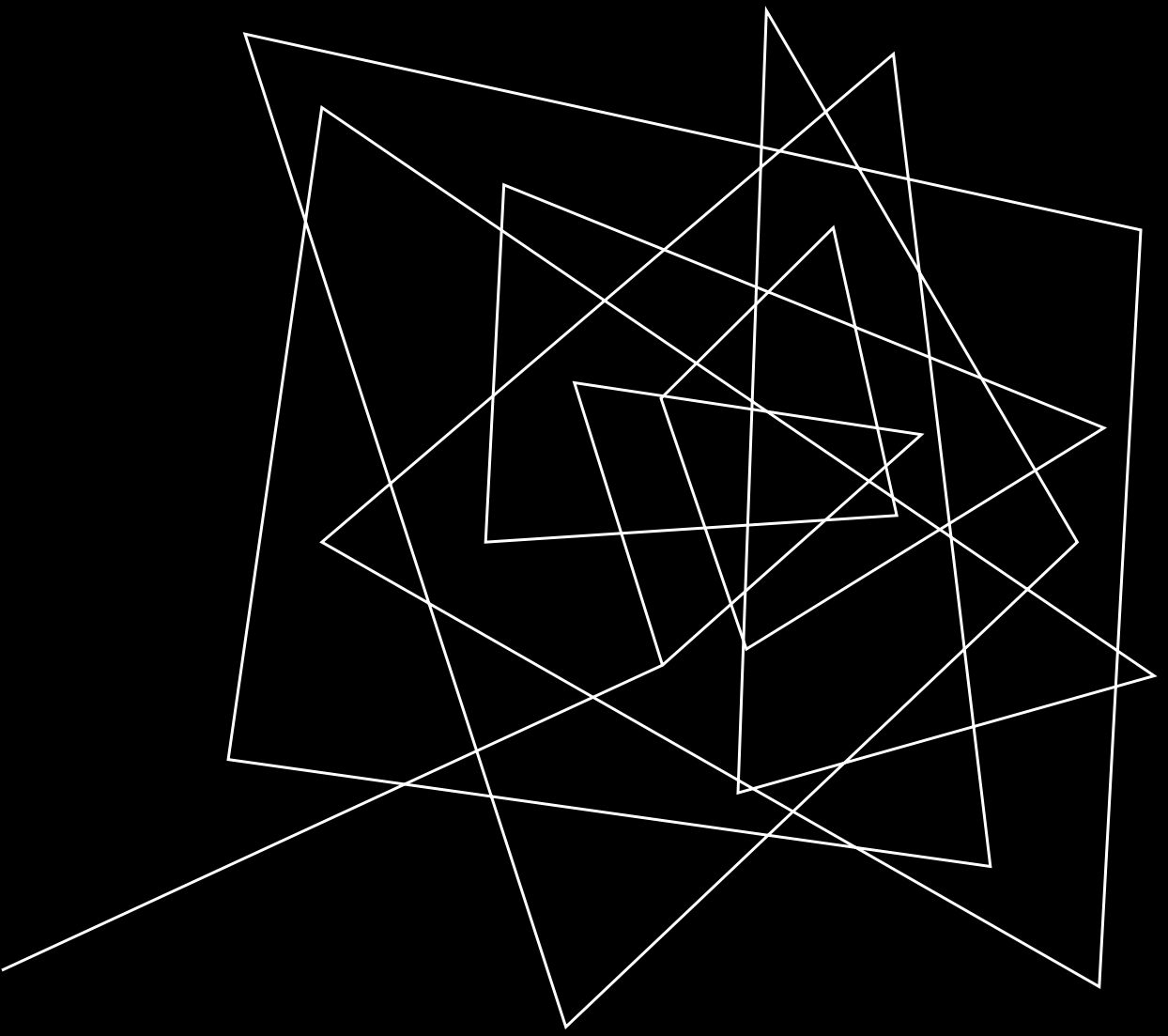
EXPLORATORY DATA ANALYSIS

TARGET VARIABLE



PREDICTORS





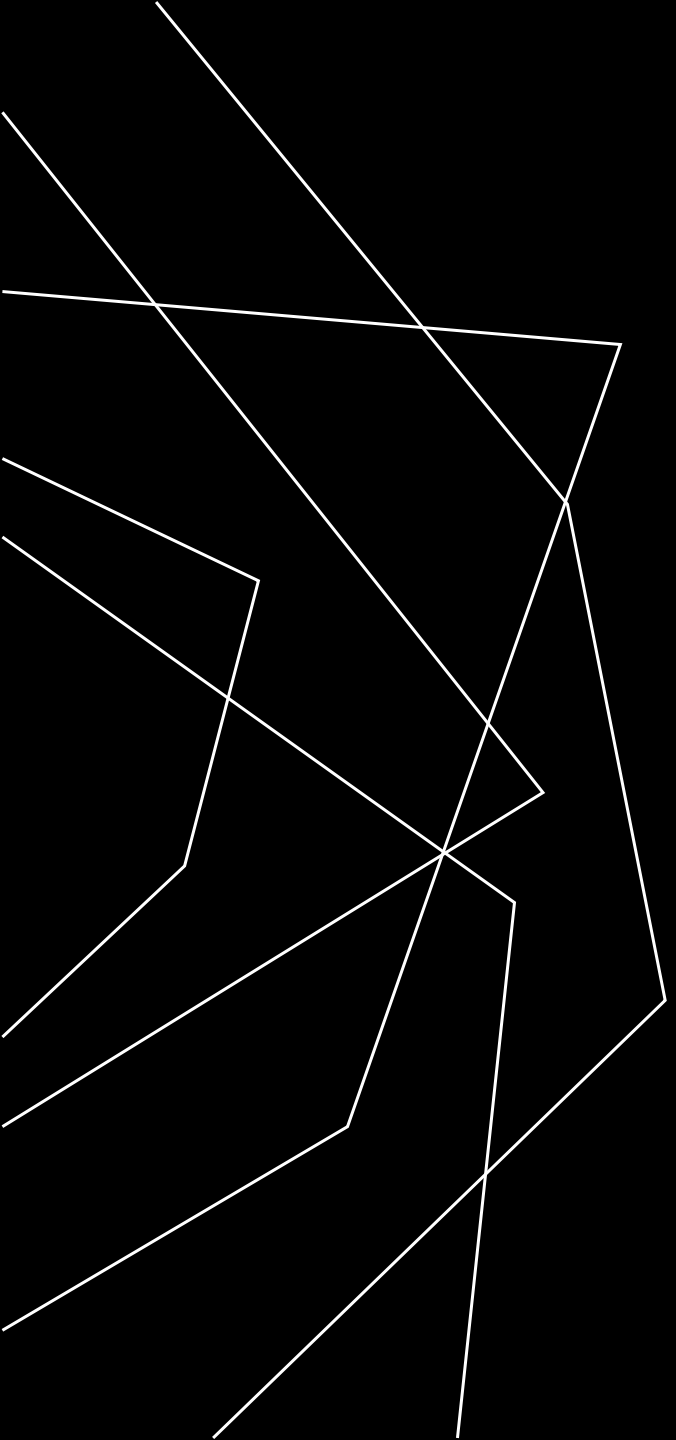
RESULTS

MODEL COMPARISON

MODEL	ACCURACY	SENSITIVITY	SPECIFICITY	AUC
Logistic Regression	0.953	0.918	0.979	0.993
Logistic Regression (Reduced)	0.918	0.849	0.969	0.984
Decision Tree	0.924	0.890	0.948	0.942
Random Forest	0.918	0.863	0.959	0.992

CONCLUSION

- Overall, a well-specified logistic regression model remains a strong and reliable approach.
  - balances predictive accuracy, interpretability, and clinical applicability
- Going forward, future work could incorporate:
  - cross-validation
  - threshold optimization to prioritize sensitivity
  - regularized models to address multicollinearity
  - expanding the dataset
  - validating results on external populations



THANK YOU

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Andreina Arias, and Stephanie Chiang

For our code please check out the project's Github .

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